6.6 Unions of Subgroups

WORKSHEET

The question

STEP II 1988, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q9

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If K is a subgroup and $a \in K$, but $b \notin K$, can ab be in K?

STAGE 1 Two subgroups cannot cover

Covers: two subgroups

Ask yourself Suppose neither G_1 nor G_2 is all of G. Choose $a \notin G_1$ and $b \notin G_2$. Where must a and b be?

Pause and try Now look at ab. It lies in G_1 or in G_2. In each case, write a or b as a product of elements of that subgroup.

STAGE 2 **Three subgroups can**

Covers: three subgroups

Pause and try Try a small group in which every element is its own inverse. What are its subgroups of two elements?

What's really going on

The language of symmetry

Groups arose in the nineteenth century from several directions at once: Galois's study of permutations of the roots of polynomials (Section 6.4), Cauchy's work on permutations, and the study of symmetries in geometry and crystallography. The abstract definition above, in which a group is anything satisfying the three axioms, was settled by the end of the century. Its power is generality: a theorem proved from the axioms, like the one in this question, holds simultaneously for groups of numbers, of matrices, of symmetries and of permutations. Today groups are everywhere: in particle physics (the standard model is built on symmetry groups), in chemistry (molecular symmetry determines spectra), in cryptography (elliptic curve groups, Section 1.9) and in the solution of Rubik's cube.

How many subgroups does it take?

The question suggests a natural problem: what is the smallest number of proper subgroups whose union is the whole group? It is never 2, by the question. In 1926 Gaetano Scorza showed that a group is the union of three proper subgroups exactly when it has the Klein four-group as a "quotient". In 1994 J. H. E. Cohn studied the general problem and conjectured several results, and in 1997 M. J. Tomkinson proved that there is no group for which this smallest number is seven, although every number from 3 to 6 occurs. Questions like this, simple to state and surprisingly subtle, are typical of group theory.

Vector spaces

A similar argument shows that a vector space over the real numbers (such as the plane, or three-dimensional space) can never be the union of finitely many proper subspaces: a plane is not the union of finitely many lines through the origin. But over a finite field it can be: the plane over the field with two elements, which is just the Klein four-group again, is the union of its three lines.

Try this next

1. **Check the group.** Write out the multiplication table of the Klein four-group and check the group axioms. Show that it is not the same as the group $\{1, \mathrm{i}, -1, -\mathrm{i}\}$ under multiplication. (*Hint:* which elements are their own inverses?)

2. **Cyclic groups.** A group is *cyclic* if every element is a power of a single element g. Show that a cyclic group is never the union of proper subgroups, however many. (*Hint:* which subgroup contains g?)

3. **The whole numbers.** Show directly that $(\mathbb{Z}, +)$ is not the union of the subgroups $2\mathbb{Z}$ (even numbers) and $3\mathbb{Z}$ (multiples of 3). Is it the union of any two proper subgroups?

4. **Lines in the plane.** HARDERShow that the plane $\mathbb{R}^2$ is not the union of finitely many lines through the origin. (*Hint:* each line contains at most one point of the parabola $y = x^2$ other than the origin.)

Further reading

- **M. A. Armstrong, *Groups and Symmetry*** (Springer). A beautifully illustrated first course, driven by geometric examples.
- **Nathan Carter, *Visual Group Theory*** (Mathematical Association of America). Groups through pictures, ideal before university.
- **Peter Neumann, Gabrielle Stoy & Edward Thompson, *Groups and Geometry*** (Oxford University Press). A classic Oxford first-year text.

The full guided solution is in Section 6.6:
https://beyondstep.keithmcnulty.org/ch06/subgroup-unions.html